A Transferable Reinforcement Learning Framework for Option Price Forecasting: Enhancing Rainbow with Two-Stage Adaptation

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ABSTRACT

Accurate option price forecasting is hindered by stochastic volatility, discontinuous jumps, pronounced non-stationarity, and scarce high-quality data. We propose a transferable reinforcement-learning framework that augments Rainbow with a two-stage adaptation strategy. In the pre-training stage, synthetic option time-series are generated via an SVJJ (stochastic volatility with jumps in price and volatility) model that embeds long-range dependence (Hurst=0.68) and micro-structure heterogeneity (R-factor). In the fine-tuning stage, the agent adaptsthrough transfer learningto real CSI 300 ETF option data, achieving fast convergence under noisy conditions. Three targeted upgrades further strengthen Rainbow: KL-divergence-based prioritized replay, dynamic learning-rate adjustment, and Bayesian neural layers for uncertainty estimation. Against 18 baselines classical regressors, deep networks, and leading RL methodsthe full Rainbow + variant delivers an MSE 7.83, MAE 1.66, Rš 0.97, reducing error 35 % vs. standard Rainbow and >50 % vs. the best deep-learning baseline. Over 90 % of call/put price errors fall below 0.15, and trading simulations show a 14.2 % lower maximum drawdown with a 22 % Sharpe-ratio uplift. The results confirm that coupling model-driven synthetic data with an adaptively trained, uncertainty-aware Rainbow agent provides a reliable, high-precision tool for option price forecasting and risk-adjusted decision support.

1. Introduction

Modeling the price behavior of financial assets remains a longstanding challenge due to the complex and volatile nature of financial markets. Financial time series are typically nonlinear, non-stationary, and subject to stochastic volatilityproperties driven by a range of factors, including macroeconomic trends, policy changes, investor sentiment, and capital flow dynamics. These interacting variables often lead to sharp, irregular price movements that are difficult to predict and interpret using conventional modeling techniques. Figure 1 illustrates the volatility comparison across major financial assets from 2020 to 2025, highlighting the diverse patterns of price fluctuations across different asset classes.

In particular, the "depth" of price fluctuations encompasses not only the magnitude and frequency of changes within a specific period but also underlying structural signals within the data, such as shifts in momentum, changes in volatility regimes, and variations in market liquidity. To capture these patterns, models must be capable of handling both continuous dynamics and discrete jumps, influenced by a combination of observable and latent variables.

Machine learning and deep learning algorithms have seen widespread adoption in financial forecasting due to their capacity to process high-dimensional data and uncover underlying patterns. In this study, we investigate several classical regression modelsincluding Ordinary Least Squares (OLS), Lasso, Ridge Regression, and Elastic Netas baseline benchmarks. These linear models are particularly useful for feature selection and scenarios where interpretability is crucial, offering both computational efficiency and ease of implementation (Gao et al., 2025). However, their reliance on linear assumptions limits their ability to capture the complex, nonlinear dynamics often found in financial time series. Recent advances in combining multi-view data features and machine learning approaches have shown promise in improving stock prediction accuracy (Yang et al., 2024), while belief rule-based expert systems have demonstrated effectiveness in financial forecasting tasks (Hossain et al., 2022).

Linear models exhibit well-known limitations in capturing nonlinear relationships. In contrast, tree-based methods, such as Decision Trees and Random Forests, provide distinct advantages in modeling complex interactions and quantifying variable importance. While effective in moderately complex settings, their

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performance may degrade in the presence of high non-stationarity, a feature commonly observed in financial market data (Ozcalici & Bumin, 2024).

Additionally, deep learning techniquesincluding Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, and Transformer-based architecturesare employed to model intricate temporal dependencies and long-range correlations. Figure 2 provides an architectural comparison of traditional machine learning and deep learning models, illustrating the increasing complexity and expressiveness from simple regression models to sophisticated neural networks. Although originally developed for computer vision tasks, CNNs have proven effective in extracting localized features, which can be leveraged for financial time series modeling (Chen et al., 2024a). RNNs and LSTMs, designed specifically for sequential data, are adept at preserving temporal context, which is critical for accurate time-dependent predictions (Sun et al., 2024). Gated Recurrent Unit (GRU) architectures have also demonstrated strong performance in stock index prediction tasks, offering computational efficiency while maintaining predictive accuracy (Gupta et al., 2022). More recently, Transformer architectures, which rely on self-attention mechanisms, have shown advantages in modeling global dependencies while enabling efficient parallel computation, making them increasingly attractive for financial forecasting applications (Heednacram et al., 2024).

Despite their powerful modeling capabilities, deep learning models often require large amounts of labeled data and substantial computational resources. Moreover, the opaque nature of their decision-making processes raises concerns about interpretability and transparency, which are particularly important in financial contexts where model outputs can influence high-stakes decisions.

Classical econometric models have long served as a foundation for analyzing financial time series. However, their assumptions often fail under high-volatility or discontinuous market conditions. In contrast, reinforcement learning (RL) offers a flexible, data-driven approach to sequential decision-making, particularly in uncertain and dynamic environments. In this study, we examine several widely-used RL algorithms, including Deep Q-Networks (DQN), Proximal Policy Optimization (PPO), Twin Delayed Deep Deterministic Policy Gradient (TD3), Actor-Critic (AC), and Deep Deterministic Policy Gradient (DDPG). These algorithms conceptualize the interaction between an intelligent agent and a dynamic market environment as a Markov Decision Process (MDP), where the agent learns to optimize its actions in order to maximize cumulative rewards, while accounting for uncertainty and temporal dependencies (Kashif et al., 2022). Figure 3 illustrates the agentenvironment interaction loop in which market states serve as observations, trading actions represent agent decisions, and portfolio returns act as rewards.

The DQN algorithm adopts a value-based approach, using deep neural networks to approximate Q-values for discrete actions. However, it often suffers from overestimation bias and instability when applied to continuous action spaces, which are common in financial settings. Policy gradient methods such as PPO and AC address these limitations by directly optimizing the policy, resulting in improved convergence behavior and robustness. Meanwhile, algorithms like DDPG and TD3, which combine actor and critic components, are tailored for continuous control tasks and have demonstrated better sample efficiency and performance in high-dimensional financial environments (Canyakmaz et al., 2022; Aras, 2021).

Despite the growing effectiveness of these algorithms, their real-world deployment in financial markets remains constrained by a number of practical challenges. Financial data is often characterized by sparse and delayed reward signals, non-stationary patterns, and noisy or partially observable information. These characteristics can significantly impede the training stability and generalization capabilities of RL agents. Furthermore, issues such as reward shaping, explorationexploitation trade-offs, and sensitivity to hyperparameter tuning present additional barriers to implementation. Careful model calibration and robust design strategies are therefore essential for ensuring the practical applicability and reliability of RL-based financial systems (Soleymani & Paquet, 2021).

To address these challenges, this paper proposes a hybrid modeling framework that combines an enhanced reinforcement learning algorithm with a market simulation environment driven by a stochastic volatility jump-jump (SVJJ) model. Figure 4 illustrates the structure of the SVJJ model, highlighting how stochastic volatility and jump components interact to capture complex market dynamics. The model integrates both return-based and volatility-based jump processes, incorporating rough stochastic structures through the Hurst exponent to simulate long-term memory. On the algorithmic side, the reinforcement learning component is based on a modified Rainbow framework, augmented with three key enhancements: KL-divergence-based experience prioritization, dynamic step-size adjustment, and Bayesian parameter estimation for uncertainty calibration. These design choices aim to create a more adaptable and stable learning system, capable of responding to shifts in market regimes and noisy, unstable training signals.

To the best of our knowledge, no study has yet integrated the SVJJ model with reinforcement learning. By combining both the SVJJ model and reinforcement learning, our framework leverages the strengths of each approach. Financial environments are inherently noisy, high-dimensional, and often lack clear reward signals.

Incorporating a volatility-aware environment, such as the SVJJ model, into the learning loop can enhance realism and foster more robust policy development. Through this synergy of modeling and learning enhancements, our proposed framework aims to provide more accurate, stable, and interpretable results for complex financial forecasting tasks.

The key contributions of this paper are as follows:

1. Transferable RL Framework: This paper presents a transferable reinforcement learning framework that integrates the SVJJ model with a two-stage adaptation strategy, utilizing synthetic data generation and transfer learning to improve adaptability and generalization in complex financial environments.

2. Enhancements to the Rainbow Algorithm: We introduce three key enhancements to the Rainbow reinforcement learning algorithm: KL-divergence-based prioritized replay, dynamic learning-rate adjustment, and Bayesian neural layers for uncertainty estimation, improving stability, convergence speed, and robustness.

3. Improved Predictive Accuracy: Extensive experiments and backtesting demonstrate superior predictive accuracy and risk-adjusted performance, achieving significant reductions in prediction error and improved trading strategy outcomes compared to traditional methods and other state-of-the-art models.

The paper is organized as follows:

The remainder of this paper is structured as follows: Section 2 reviews the relevant literature. Section 3 outlines a two-stage adaptation strategy (SVJJ-driven synthetic pre-training and transfer-learning fine-tuning) alongside three Rainbow upgrades (KL-based replay, dynamic learning rates, Bayesian layers). Section 4 reports experimental results covering performance metrics, ablation studies, feature importance, and trading simulations. Finally, Section 5 concludes with practical implications.

2. Related Work

This section reviews the existing literature on option price forecasting, highlighting key contributions in reinforcement learning and stochastic volatility modeling.

2.1. Traditional econometric & ML models for option pricing

To address the limitations associated with individual reinforcement learning components, this study adopts the Rainbow algorithm, an enhanced extension of the Deep Q-Network (DQN) framework that integrates several key innovations into a unified architecture. By combining double Q-learning, dueling network architectures, prioritized experience replay, multi-step learning, distributional reinforcement learning, and noisy networks, Rainbow offers a more robust and sample-efficient learning paradigm suitable for complex environments (Lee et al., 2021).

Each component of the Rainbow algorithm contributes to improved learning stability and generalization. Double Q-learning mitigates overestimation bias by decoupling the action selection and evaluation steps, while the dueling network architecture separately estimates state values and advantages, thereby enhancing learning efficiency. Prioritized experience replay improves sample selection by emphasizing transitions with higher learning potential, and multi-step return estimation enriches the temporal context of reward propagation. Distributional reinforcement learning captures the full return distribution rather than a single expected value, allowing the model to better represent uncertainty. Noisy networks further enhance exploration by incorporating learnable noise directly into network parameters.

The integration of these mechanisms into a single framework enables the Rainbow agent to overcome the training instability and limited adaptability often observed in standard RL algorithms, especially under volatile and noise-prone conditions prevalent in financial markets. In this study, the Rainbow-based agent is trained to dynamically adjust portfolio weights by leveraging historical price information. Its objective is to optimize the long-term returnrisk profile while adhering to realistic market constraints.

Empirical results indicate that, relative to traditional learning algorithms and single-policy reinforcement learning models, the proposed Rainbow-based framework demonstrates superior performance in terms of cumulative returns, volatility control, and drawdown minimization. These outcomes reflect its enhanced capability to model both the depth and complexity of financial price fluctuations.

2.2. Deep learning for financial forecasting (CNN, LSTM, Transformer)

Transfer learning has emerged as an effective approach to enhance model generalization in data-constrained or heterogeneous environments and has been increasingly applied in financial research to mitigate issues related to limited data availability and domain variability (Salisu et al., 2022). Fundamentally, transfer learning enables a model trained in a source domain to be reused or fine-tuned in a target domain, thereby improving predictive accuracy while reducing training time and computational overhead.

In the context of financial markets, the high volatility and noise inherent in real-time data often result in overfitting and poor generalization when models are trained solely on limited empirical samples. To address this, the present study proposes a two-stage transfer learning framework designed to bridge the gap between simulated and real-world financial data.

In the first stage, a pre-training process is conducted using source domain data generated from a theoretical simulation modelan enhanced Hurst-SVJJ model that incorporates an R-factor derived from the R-measure of market microstructure information. This synthetic model is designed to replicate key statistical properties observed in financial markets by integrating long-range dependence through the Hurst exponent, stochastic volatility processes, and discrete jump dynamics. The R-factor serves as an informational increment, enriching the simulation with heterogeneous behavioral characteristics and reflecting investor-driven microstructural variations (Kirisci & Yolcu, 2022).

Following pre-training, the model is fine-tuned on target domain data consisting of empirical financial time series. This fine-tuning stage enables the model to adapt to realistic temporal patterns and noise structures while retaining knowledge acquired from the simulated environment. The proposed hybrid strategy facilitates smoother convergence, improves generalization performance, and enhances robustnessparticularly in settings with sparse or noisy observational data.

Nevertheless, the effectiveness of transfer learning in this context is contingent upon the compatibility between the synthetic and empirical domains. Disparities in statistical distributions, latent structures, or volatility regimes can result in negative transfer effects, undermining model performance. To mitigate these risks, careful design of the simulated environment and the adoption of gradual fine-tuning procedures are essential for ensuring model stability and maximizing the utility of transferred representations.

2.3. Reinforcement learning in finance (DQN Rainbow, PPO, TD3)

Although the Rainbow algorithm consolidates several state-of-the-art reinforcement learning techniques into a unified framework, its performance can still be limited in the context of financial markets, where non-stationarity, delayed reward signals, and inherent stochasticity present substantial modeling challenges (Lee et al., 2021). To enhance the algorithm's effectiveness in such environments, this study introduces several targeted improvements aimed at increasing sample efficiency, training stability, and decision robustness.

One enhancement involves modifying the prioritized experience replay mechanism. In the standard approach, transition priority is determined by the temporal-difference (TD) error; however, this scalar measure may inadequately capture learning potential in environments characterized by complex, multimodal return distributions. To overcome this limitation, we propose using the KullbackLeibler (KL) divergence between the predicted and target return distributions as the prioritization criterion. This distribution-based metric allows for a more nuanced assessment of learning significance by quantifying discrepancies in the full return distribution, thereby enabling the agent to concentrate on transitions with higher informational value and accelerating policy convergence (Bhavsar et al., 2021).

Another improvement focuses on learning rate adaptation. Rather than employing a fixed step size throughout the training process, we implement a dynamic adjustment mechanism that modulates the learning rate in response to changes in the estimated state value function. By monitoring the magnitude of value updates, this approach allows the algorithm to balance short-term and long-term rewards more effectively, particularly in the presence of abrupt market regime shifts or sentiment-driven volatility. This adaptive step-size strategy enhances the responsiveness and resilience of the learning process in real-time financial scenarios.

Additionally, to improve parameter robustness and capture epistemic uncertainty, we incorporate Bayesian neural network (BNN) techniques into the architecture. In this formulation, network parameters are modeled as probability distributions rather than fixed point estimates. This probabilistic treatment enables the model to express uncertainty through parameter variance, reducing the risk of overfitting and improving generalization to unseen or highly variable market conditions (Abdelkawy et al., 2021).

Together, these enhancements significantly increase the adaptability and reliability of the Rainbow agent when deployed in complex, high-noise financial environments. Empirical evaluation indicates that the improved algorithm exhibits greater stability, reduced regret, and superior risk-adjusted performance relative to both standard Rainbow and other baseline methods. These results underscore the importance of domain-informed algorithmic refinement in advancing reinforcement learning applications in finance.

2.4. Synthetic data & transfer learning in financial domains

Financial markets are characterized by data scarcity, non-stationarity, and high dimensionality, presenting significant challenges for traditional machine learning approaches that rely heavily on abundant high-quality training data. The inherent volatility and irregular patterns in financial time series often result in insufficient data for robust model training, particularly when attempting to capture rare market events or extreme volatility regimes that are crucial for accurate option pricing. To address these fundamental challenges, recent advances have proposed innovative frameworks that combine synthetic data generation with transfer learning methodologies to enhance model performance in financial applications.

Chen et al. (2025) propose a groundbreaking transfer learning framework that integrates structural model constraints into machine learning models by leveraging synthetic data generation followed by real-data fine-tuning. Their approach demonstrates that even misspecified structural models can effectively guide machine learning model training through a two-stage process: first pre-training neural networks on synthetic data generated from theoretical models such as the Black-Scholes framework, then fine-tuning these networks on empirical data using significantly smaller learning rates. This methodology addresses the fundamental trade-off between the interpretability of structural models and the flexibility of machine learning approaches, showing that the transfer learning model achieves a Black-Scholes implied volatility median absolute error (BSIV-MAE) that is only 32.4% of the deep learning baseline and 9.3% of the Heston model. The effectiveness of this approach is particularly pronounced for medium and long-term options, out-of-the-money contracts, and higher liquidity scenarios, demonstrating that structural constraints become more valuable when standard machine learning methods struggle to generalize beyond training data boundaries or handle underlying data generating process instabilities.

Complementing this transfer learning approach, Chen, Didisheim, and Scheidegger (2023) introduce the concept of "deep surrogate models" that utilize deep neural networks to create high-accuracy approximations of structural models, significantly accelerating model evaluation and estimation by several orders of magnitude. Their deep surrogate methodology enables computationally intensive analyses that would otherwise be prohibitive, such as high-frequency model re-estimation for constructing option-implied tail risk indices and systematic out-of-sample performance testing of structural models. The deep surrogate approach demonstrates that neural networks can accurately inherit economic constraints implied by structural models while maintaining computational efficiency, with their Bates model surrogate achieving remarkable approximation accuracy using a network with 7 hidden layers and nearly one million parameters trained on 10^6 synthetic samples. This methodology proves particularly valuable for applications requiring frequent model evaluations, such as dynamic parameter estimation and risk management scenarios, where the surrogate model reduces parameter estimation time by several orders of magnitude while preserving the theoretical foundations of the underlying structural model.

The integration of these two methodological advancestransfer learning with synthetic pre-training and deep surrogate modelingrepresents a paradigm shift in financial modeling that addresses both the data scarcity problem and the computational complexity challenges inherent in modern financial applications. Chen et al.'s transfer learning framework demonstrates that synthetic data generation can provide informative priors similar to Bayesian VAR approaches, but with enhanced capability for capturing nonlinear features and more flexible mechanisms for determining "prior strength" through hyperparameter tuning rather than fixed sample size ratios. Meanwhile, the deep surrogate approach enables the practical implementation of high-dimensional structural models in real-time applications, facilitating the construction of time-varying parameter instability measures and their correlation with market liquidity conditions. These methodological innovations collectively show that the combination of model-driven synthetic data generation with adaptively trained, uncertainty-aware learning systems provides a reliable pathway for enhancing both prediction accuracy and economic interpretability in financial forecasting tasks.

3. Methodology

This study estimates historical volatility by computing the standard deviation of the financial product's historical price series, providing a quantitative measure of price fluctuation over a specific time horizon. Additionally, expected volatility is assessed through the analysis of implied volatility derived from options market data, which reflects market participants' collective expectations regarding future price dynamics.

Implied volatility, as inferred from option pricing models, serves as a forward-looking indicator that encapsulates anticipated market uncertainty. It offers valuable guidance for the formulation of trading strategies and risk management decisions. The SVJJ model, with its capacity to capture both stochastic volatility and discontinuous jumps in asset price trajectories, provides a more comprehensive description of complex financial behaviors. Meanwhile, reinforcement learning enhances model adaptability through iterative optimization and policy refinement, allowing the agent to respond to evolving market regimes.

The integration of these two components addresses the limitations inherent in traditional time-series models, particularly under extreme or non-stationary conditions, while also compensating for the weaknesses of reinforcement learning when applied to high-dimensional financial data. This hybrid approach leverages the complementary strengths of both techniques, contributing to improved modeling fidelity and robustness. The experimental findings suggest that the proposed method holds potential for achieving more favorable returnrisk trade-offs, offering meaningful implications for practical financial decision-making.

3.1. SVJJ Data-Generation Engine

The problem is basically formulates as a **Markov Decision Process (MDP)**. An MDP is defined by a tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$, where:

- **State Space $\mathcal{S}$:** The state $s_t \in \mathcal{S}$ at time t is a comprehensive representation of the market environment. It includes a lookback window of historical option and underlying asset prices, time to maturity τ, moneyness K/S_t, and other technical indicators derived from the market data.

- **Action Space $\mathcal{A}$:** The action $a_t \in \mathcal{A}$ represents the agent's decision at each step. In our framework, this corresponds to determining the proportion of capital to allocate among a set of assets, such as a specific call option, a put option, and a risk-free cash position. The action is a vector $a_t = [w_{\text{call}}, w_{\text{put}}, w_{\text{cash}}]$ where $\sum w_i = 1$.
- **Transition Probability P:** $P(s_{t+1}|s_t, a_t)$ defines the probability of transitioning to state s_{t+1} after taking action a_t in state s_t. In financial markets, this function is unknown and is implicitly defined by the market dynamics. Our agent learns to operate within this environment without explicit knowledge of P.
- **Reward Function R:** The reward r_{t+1} measures the immediate benefit of taking action a_t in state s_t. We define it as the risk-adjusted portfolio return, commonly the Sharpe ratio over the decision period, to encourage strategies that balance profitability and risk.
- **Discount Factor γ:** The discount factor $\gamma \in [0, 1)$ balances the importance of immediate versus future rewards.

The agent's goal is to learn a policy $\pi(a|s)$ that maximizes the expected discounted cumulative reward:

$$G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}.$$

We formally model the problem as a **Markov Decision Process (MDP)** in order to apply reinforcement learning to option price forecasting and trading.

The **Stochastic Volatility with Jumps in both price and volatility (SVJJ)** model is designed to replicate key statistical properties observed in financial markets by integrating long-range dependence through the Hurst exponent, stochastic volatility processes, and discrete jump dynamics. The continuous-time dynamics of the asset price S_t are governed by:

$$\frac{dS_t}{S_t} = (\mu - \lambda\bar{k})dt + \sqrt{V_t}dW_t^{(1)} + k_t dN_t$$

where μ is the drift, V_t is the stochastic volatility, $W_t^{(1)}$ is a standard Wiener process, N_t is a Poisson process with intensity λ governing the arrival of jumps. The jump size k_t is typically log-normally distributed. The volatility process V_t follows a mean-reverting square root process, itself subject to jumps. This simulation engine allows us to generate a rich dataset of realistic price trajectories for robust pre-training.

We pre-train our agent in a synthetic environment which is formed by SVJJ in order to address the data scarcity and non-stationarity challenges inherent in financial markets.

3.2. Enhancements to the Rainbow Algorithm

The standard Rainbow algorithm refers to a combination of six key enhancements to Deep Q-Networks (DQN) which integrates distributional RL, prioritized experience replay, and other improvements to better handle the stochastic and non-stationary nature of financial markets.

While the standard Rainbow algorithm provides a strong baseline, we introduce three targeted enhancements to optimize its performance for financial forecasting, as shown in Figure 7

3.2.1. KL-Divergence Prioritized Experience Replay

Standard Prioritized Experience Replay (PER) uses the magnitude of the TD-error as the priority for sampling transitions. However, this scalar value may not fully capture the "surprise" in an update, especially for distributional RL. We replace the TD-error with the Kullback-Leibler (KL) divergence between the predicted and target return distributions.

Rainbow models the value function as a distribution over a set of discrete supports $\{z_i\}_{i=1}^N$. Let the predicted distribution for state-action (s, a) be $p_\theta(s, a)$ with probabilities $\{p_i\}_{i=1}^N$ for each atom z_i. The target distribution $p_{\theta'}(s', a')$ is projected onto the same supports. The priority of a transition j is then calculated as:

$$p_j = D_{\text{KL}}\left(p_{\text{target}} \parallel p_{\text{predict}}\right) = \sum_{i=1}^{N} p_{\text{target},i} \log\left(\frac{p_{\text{target},i}}{p_{\text{predict},i}}\right) \tag{1}$$

This allows the agent to focus on transitions where its belief about the entire return distribution was most incorrect, leading to more efficient learning.

3.2.2. Dynamic Step-Size Adjustment

Financial markets exhibit shifting volatility regimes. A fixed learning rate (α) can be suboptimal, leading to instability during high volatility or slow convergence in stable periods. We implement a dynamic step-size adjustment mechanism linked to the model's uncertainty. Using the Bayesian layers (described next), we can estimate the epistemic uncertainty $\sigma(s, a)$ of the Q-value prediction. The learning rate for a given update is then modulated as:

$$\alpha_t = \frac{\alpha_0}{1 + \beta \cdot \sigma_t(s, a)} \tag{2}$$

where α_0 is the initial learning rate and β is a scaling factor. This heuristic reduces the learning rate when the agent is uncertain about its value estimates, preventing large, potentially destabilizing updates.

3.2.3. Bayesian Neural Network Layers

A Bayesian Neural Network (BNN) is a probabilistic variant of traditional neural networks where the weights are represented as probability distributions rather than deterministic values. To explicitly model and quantify epistemic uncertainty, we replace the standard linear layers in the network with Bayesian neural network (BNN) layers. Instead of learning point estimates for the network weights W, we learn posterior distributions $p(W|D)$ over them, where D is the experienced data.

As the true posterior is intractable, we use variational inference to approximate it with a tractable distribution $q_\theta(W)$, typically a Gaussian parameterized by its mean and variance $\theta = (\mu, \rho)$. We then minimize the KL-divergence between the approximate and true posteriors, which is equivalent to minimizing the negative Evidence Lower Bound (ELBO). The loss function becomes:

$$\mathcal{L}(D, \theta) = \underbrace{D_{\mathrm{KL}}\big(q_\theta(W) \,\|\, p(W)\big)}_{\text{Complexity Cost}} - \underbrace{\mathbb{E}_{q_\theta(W)}[\log p(D|W)]}_{\text{Likelihood Cost}} \tag{3}$$

The complexity term acts as a regularizer, penalizing deviations from the prior $p(W)$, while the likelihood term encourages fitting the data. This allows the agent to express its confidence in its predictions, which is crucial for risk management in finance.

3.3. Algorithm Pseudocode

The complete training process of our enhanced Rainbow agent (R-Full) is summarized in Algorithm 1. This algorithm outlines the two-stage adaptation strategy, beginning with pre-training on the SVJJ simulation engine and concluding with fine-tuning on real market data.

3.4. Two-Stage Adaptation Framework

Our core methodological contribution is a two-stage transfer learning framework designed to bridge financial theory with real-world market noise.

As illustrated in Figure 6, the first stage involves pre-training the Rainbow agent on a large volume of synthetic data generated by the SVJJ engine. This allows the agent to learn fundamental trading strategies and value function approximations in a stable environment. In the second stage, the learned weights from the pre-trained agent are used to initialize the model, which is then finetuned on the actual, often noisy, CSI 300 ETF option data. This transfer learning step significantly accelerates convergence and improves generalization.

4. Experimental Setup

4.1. Data and Summary Statistics

We use CSI 300 ETF options daily data from the Shanghai Stock Exchange over the period from 2020 to 2025. The dataset encompasses both call and put options with various strike prices and expiration dates, providing a comprehensive view of the Chinese options market dynamics. The descriptive statistics of our main variables are shown in Table 1.

To handle missing values in our dataset, we employ the closing price of the previous trading day for option prices. For missing historical volatility and risk-free rates, we apply linear interpolation methods to ensure data continuity. In terms of outlier processing, we remove observations with realized excess returns and model-based expected returns exceeding the 99% quantile from the entire sample to mitigate the impact of extreme market events and ensure robust model training.

The dataset preprocessing also includes standardization of key features such as moneyness, time to maturity, and volatility measures to improve model convergence. We further apply a rolling window approach for calculating historical volatility using a 30-day lookback period, which aligns with standard market practice for short-term volatility estimation.

4.2. Baseline Models

To rigorously assess the predictive performance of our proposed model, we compare it against a diverse suite of 18 baseline approaches, encompassing traditional machine learning algorithms, deep learning architectures, and reinforcement learning methods. The inclusion of such a wide array of benchmarks ensures that the evaluation is comprehensive and reflective of the modeling strategies commonly adopted in quantitative finance. Within the traditional paradigm, we select six representative models, including linear and regularized regressions (e.g., OLS, Lasso, Ridge, and Elastic Net) as well as tree-based methods such as Decision Trees and Random Forests. These models are widely used in the literature and offer interpretability and robustness under different data regimes. In contrast, the deep learning category includes four state-of-the-art architectures: CNNs and RNNs for sequence learning, LSTMs for modeling long-term dependencies, and Transformers for capturing global attention structures. Reinforcement learning baselines include both value- and policy-based agents (e.g., DQN, PPO, TD3, DDPG, and AC), alongside the Rainbow algorithm, which serves as a direct point of comparison for our enhancements. All models are evaluated under identical data preprocessing and training configurations to ensure fairness and consistency.

Table 1
Summary Statistics of CSI 300 ETF Options Dataset

	Mean	Sd	Min	Q10	Q25	Median	Q75	Q90	Max	Skew	Kurt
A: All Options ($n = 156,847$)											
Realized excess return	-0.18	1.09	-1.12	-0.98	-0.96	-0.92	0.42	1.38	6.12	1.94	4.22
Model-based expected return	0.92	3.28	-1.08	-0.95	-0.79	-0.09	1.18	3.67	28.43	3.97	20.15
Option maturity	245.18	218.76	12.00	32.00	95.00	158.00	372.00	553.00	735.00	1.08	0.05
Option moneyness	1.02	0.14	0.43	0.87	0.96	1.02	1.07	1.16	3.12	0.52	4.18
Implied volatility	0.22	0.08	0.04	0.13	0.16	0.21	0.26	0.32	1.35	1.48	4.89
Historical volatility	0.19	0.11	0.03	0.09	0.12	0.17	0.24	0.31	1.42	2.56	11.23
Option delta	0.02	0.58	-0.88	-0.73	-0.48	-0.08	0.52	0.77	0.92	0.03	-1.41
B: Call Options ($n = 77,234$)											
Realized excess return	0.07	1.18	-1.12	-0.98	-0.96	-0.15	0.76	1.67	6.12	1.52	2.68
Model-based expected return	1.84	4.15	-1.08	-0.92	-0.51	0.43	2.15	6.02	28.43	3.18	11.47
Option maturity	241.67	216.82	12.00	32.00	95.00	158.00	372.00	553.00	735.00	1.12	0.14
Option moneyness	1.01	0.13	0.43	0.86	0.95	1.01	1.07	1.15	2.04	0.04	2.73
Implied volatility	0.22	0.08	0.06	0.13	0.16	0.21	0.26	0.32	1.35	1.42	4.35
Historical volatility	0.19	0.11	0.03	0.09	0.12	0.17	0.24	0.31	1.42	2.58	11.52
Option delta	0.53	0.26	0.12	0.17	0.32	0.52	0.72	0.87	0.92	-0.06	-1.21
C: Put Options ($n = 79,613$)											
Realized excess return	-0.42	0.97	-1.12	-0.98	-0.97	-0.96	-0.31	0.91	6.12	2.83	8.67
Model-based expected return	0.02	1.78	-1.08	-0.96	-0.88	-0.47	0.41	1.84	28.21	4.35	30.85
Option maturity	248.59	220.58	12.00	32.00	95.00	158.00	372.00	553.00	735.00	1.05	-0.01
Option moneyness	1.02	0.14	0.54	0.87	0.96	1.02	1.08	1.18	3.12	0.89	5.02
Implied volatility	0.22	0.08	0.04	0.13	0.16	0.21	0.26	0.32	1.28	1.54	5.37
Historical volatility	0.19	0.11	0.03	0.09	0.12	0.17	0.24	0.31	1.42	2.54	10.96
Option delta	-0.48	0.26	-0.88	-0.87	-0.72	-0.48	-0.32	-0.17	-0.12	0.03	-1.23

Notes: This table reports descriptive statistics for main variables in the CSI 300 ETF options dataset over the period from 2020 to 2025. Panel A, B, and C report the statistics for all options, call options, and put options, respectively. Realized excess return represents the actual holding-to-maturity option excess return. Model-based expected return is the theoretical excess return calculated using our enhanced SVJJ-Rainbow framework. Option maturity denotes the number of calendar days until option expiration. Option moneyness is defined as the ratio between the underlying ETF price and the option strike price. Implied volatility represents the market's expectation of future volatility embedded in option prices. Historical volatility is calculated using a 30-day rolling window of underlying asset returns. Option delta measures the sensitivity of option price to changes in the underlying asset price.

4.3. Evaluation Metrics

In addition to accuracy-based evaluation, our assessment framework integrates a comprehensive set of metrics designed to capture model quality across multiple dimensions. Standard regression metrics such as MSE, RMSE, MAE, and Rš are employed to assess the precision of point forecasts, supplemented by measures like Median Absolute Error, Maximum Error, and Explained Variance to highlight error distributions and overall model fidelity. Considering the probabilistic outputs of our model, distributional metrics such as KL divergence and calibration curves are used to quantify the alignment between predicted and actual return distributions. Moreover, financial relevance is ensured through risk-adjusted performance indicators like Sharpe and Sortino ratios, as well as tail risk measures including VaR and CVaR. Simulated trading strategies incorporating real-world constraints such as transaction costs, turnover limits, and liquidity impacts are used to evaluate strategy robustness. Furthermore, statistical significance tests including paired t-tests, Wilcoxon signed-rank tests, and Diebold-Mariano tests are performed to verify the reliability of observed differences, supported by Model Confidence Set procedures to identify statistically indistinguishable models.

4.4. Implementation Details

The entire experimental setup is constructed with an emphasis on computational rigor and reproducibility. All implementations are carried out in Python 3.9 using PyTorch 1.12 and Stable-Baselines3, with training conducted on NVIDIA RTX 4090 GPUs and Intel Xeon CPUs for preprocessing. Raw CSI 300 ETF option data is cleaned, interpolated, and normalized via standardized pipelines, with volatility features computed using rolling-window statistics. The proposed Rainbow-based agent utilizes a dueling architecture with Bayesian extensions and distributional modeling over a fixed support. A two-stage training regime is adopted, beginning with synthetic data pre-training followed by fine-tuning on real data, each using carefully tuned hyperparameters and dynamic learning schedules. Grid search and Bayesian optimization are jointly used for hyperparameter selection, while consistent random seeding and model checkpointing ensure full reproducibility. Overall, the training framework supports both high computational efficiency and methodological transparency, enabling scalable deployment in real-world financial environments.

5. Results and Discussion

5.0.1. Performance Metrics: MSE and MAE

The enhanced SVJJ model demonstrates strong predictive accuracy in option pricing tasks. Figure 6 shows the scatter plots of predicted versus true values for call and put options, where the data points align closely along the 45-degree diagonal. This alignment indicates high linear correlation between predicted and actual values, confirming the model's ability to capture option price formation patterns accurately.

Figure 7 displays time series comparison for the first 50 samples, showing actual and predicted prices for both call and put options. The predicted trajectories track the true price curves closely, demonstrating the model's effectiveness in capturing temporal price dynamics. The strong alignment reflects the low MSE and MAE values achieved by the model.

The small deviations between predicted and actual prices confirm the model's reliability for option pricing. The high accuracy shown in both scatter plots and time series validates the model's precision in capturing price fluctuations, making it suitable for financial forecasting applications.

Figure 8 shows the training and validation loss curves over 50 epochs. Both curves follow a downward trend, indicating successful convergence and parameter optimization. Training loss decreases steadily, showing improved data fitting, while validation loss, despite minor fluctuations, maintains stable generalization performance. The small fluctuations suggest occasional overfitting, but the overall trend confirms effective generalization to new data.

Figure 9 presents the prediction error distribution for call and put options on the test dataset. Most absolute errors concentrate near zero, with over 90

These results confirm that the proposed model achieves excellent predictive performance in option pricing, with consistently low MSE and MAE values on both training and validation datasets. The convergence patterns and bounded prediction errors demonstrate strong generalization capability and practical value for financial forecasting.

5.0.2. Feature Importance Analysis

Figure 10 shows the relative importance of input features in the prediction framework. Moneyness ranks as the most important feature, consistent with option pricing theory where it directly influences intrinsic value and pricing behavior. Other key contributors include forward price and discount factor, which provide context for future price expectations and time-value adjustments.

Volatility shows relatively lower importance in the rankings. This may result from its implicit representation through correlated variables, especially moneyness, which captures much of the variance typically explained by volatility. Time-to-maturity and Hurst index contribute moderately, capturing temporal structure and long-memory effects in the time series. The feature importance distribution shows that the model effectively prioritizes high-impact predictors while incorporating secondary signals for stability and interpretability.

5.0.3. Comparison of Model Improvements

We conducted comprehensive performance evaluation of the proposed framework and quantified the benefits of different enhancements through three experimental configurations: (i) baseline models without transfer learning, (ii) deep learning and reinforcement learning models with and without transfer learning, and (iii) Rainbow architecture variants with transfer learning.

All models were evaluated using standard regression metrics: MSE, RMSE, MAE, Rš, Median Absolute Error, Maximum Error, and Explained Variance Score. These metrics capture both central tendency and dispersion of prediction errors, providing comprehensive model quality assessment.

Training time and computational resource consumption were also recorded to evaluate practical feasibility and efficiency. This dual evaluation covers both predictive performance and operational cost, enabling complete comparison of model applicability in real-world financial forecasting.

Performance of All Models Without Transfer Learning We established comprehensive performance baselines by evaluating modeling approaches across three categories: traditional machine learning algorithms, deep learning architectures, and reinforcement learning frameworks. Traditional models included OLS, Lasso, Ridge Regression, Elastic Net, Decision Tree, and Random Forest. Deep learning methods covered CNN, RNN, LSTM, and Transformer architectures. Reinforcement learning models included DQN, PPO, TD3, Actor-Critic, DDPG, and the original Rainbow algorithm.

Each model was trained on the same dataset without transfer learning and evaluated on a standardized test set for fair comparison. Evaluation metrics included MSE, MAE, and Rš, providing comprehensive view of prediction accuracy and generalization capacity.

Deep Learning and Reinforcement Learning Models: Transfer Learning vs. From-Scratch Training To quantify the effectiveness of the transfer learning framework introduced in Section 2.4, we conducted systematic comparative experiments on selected deep learning and reinforcement learning architectures under two distinct training protocols: standard from-scratch

Table 2
Performance Comparison of All Models (Part 1)

Model	MSE	RMSE	MAE	Rš
OLS	42.58	6.54	5.37	0.55
Lasso	41.25	6.41	5.18	0.57
Ridge	40.72	6.38	5.13	0.58
Elastic Net	40.45	6.35	5.11	0.59
Decision Tree	36.19	6.00	4.82	0.63
Random Forest	32.77	5.74	4.43	0.66
CNN	17.18	4.13	3.00	0.86
RNN	18.47	4.29	3.09	0.84
LSTM	16.09	4.01	2.91	0.87
Transformer	15.41	3.91	2.83	0.88
DQN	20.18	4.49	3.41	0.82
PPO	19.12	4.36	3.29	0.83
TD3	16.52	4.07	2.96	0.87
Actor-Critic	18.03	4.25	3.17	0.85
DDPG	17.38	4.17	3.04	0.86
Rainbow	15.01	3.88	2.76	0.89

Table 3
Performance Comparison of All Models (Part 2)

Model	Median AE	Max Error	Explained Variance
OLS	5.22	21.03	0.54
Lasso	5.08	20.24	0.56
Ridge	5.02	19.75	0.57
Elastic Net	4.93	19.61	0.58
Decision Tree	4.67	17.96	0.62
Random Forest	4.32	16.63	0.65
CNN	2.86	9.79	0.85
RNN	2.94	10.08	0.83
LSTM	2.69	9.13	0.86
Transformer	2.59	8.93	0.87
DQN	3.13	10.52	0.81
PPO	2.98	10.04	0.82
TD3	2.74	8.97	0.86
Actor-Critic	2.91	9.68	0.84
DDPG	2.84	9.43	0.85
Rainbow	2.56	8.69	0.88

initialization and transfer learning with pre-trained weights from the theoretical SVJJ simulation domain. This experimental design enables precise measurement of knowledge transfer effects across critical performance dimensions, specifically convergence efficiency, generalization capability, and prediction accuracy.

Experimental findings demonstrate that transfer learning models consistently outperform their from-scratch counterparts, achieving accelerated convergence rates and substantially reduced computational training overhead. More importantly, transferred models maintain superior predictive performance on out-of-sample data, indicating enhanced generalization across market regimes. These performance gains are particularly pronounced within reinforcement learning frameworks, where the combination of high-dimensional state

Table 4
Comparison of Models With vs. Without Transfer Learning (Part 1)

Model	Transfer Learning	MSE	RMSE	MAE
CNN	No	17.20	4.14	3.01
CNN	Yes	11.27	3.35	2.22
RNN	No	18.50	4.30	3.10
RNN	Yes	12.38	3.52	2.36
LSTM	No	16.10	4.01	2.90
LSTM	Yes	10.76	3.28	2.06
Transformer	No	15.40	3.92	2.82
Transformer	Yes	9.47	3.07	1.91
DQN	No	20.20	4.49	3.40
DQN	Yes	14.90	3.86	2.70
PPO	No	19.10	4.37	3.30
PPO	Yes	13.60	3.69	2.55
TD3	No	16.50	4.06	2.95
TD3	Yes	13.70	3.70	2.60
Rainbow	No	15.00	3.87	2.78
Rainbow	Yes	12.10	3.48	2.35

Table 5
Comparison of Models With vs. Without Transfer Learning (Part 2)

Model	Transfer Learning	Rš	Median AE	Max Error
CNN	No	0.86	2.85	9.80
CNN	Yes	0.93	1.94	6.83
RNN	No	0.84	2.95	10.00
RNN	Yes	0.91	2.11	7.48
LSTM	No	0.87	2.70	9.10
LSTM	Yes	0.94	1.84	6.37
Transformer	No	0.88	2.60	8.90
Transformer	Yes	0.95	1.68	5.78
DQN	No	0.82	3.15	10.50
DQN	Yes	0.88	2.45	8.10
PPO	No	0.83	3.00	10.00
PPO	Yes	0.89	2.25	7.60
TD3	No	0.87	2.75	9.00
TD3	Yes	0.90	2.40	7.80
Rainbow	No	0.89	2.55	8.70
Rainbow	Yes	0.92	2.10	7.20

representations and sparse reward structures typically impedes efficient learning from random initialization.

The systematic integration of transfer learning strategies yields significant improvements in both computational efficiency and model robustness, with benefits most evident in data-constrained environments characterized by limited high-quality financial datasets and elevated market volatility.

Figure 11 demonstrates the quantitative performance improvements in Mean Squared Error (MSE) reduction achieved through transfer learning implementation across diverse model architectures. The results reveal consistent performance enhancement patterns across all evaluated frameworks, with Transformer-based architectures exhibiting the most substantial

Table 6
Performance of Rainbow Variants After Transfer Learning (Part 1)

Model Variant	MSE	RMSE	MAE	Rš
R-Base	12.10	3.48	2.35	0.92
R-KL	10.82	3.29	2.17	0.93
R-DynStep	11.22	3.35	2.19	0.93
R-Bayes	9.79	3.13	1.94	0.94
R-KLStep	9.21	3.02	1.89	0.95
R-KLBayes	8.72	2.95	1.81	0.96
R-StepBayes	8.88	2.98	1.84	0.96
R-Full	7.83	2.80	1.66	0.97

Table 7
Performance of Rainbow Variants After Transfer Learning (Part 2)

Model Variant	Median AE	Max Error	Explained Variance
R-Base	2.10	7.20	0.91
R-KL	1.94	6.52	0.92
R-DynStep	1.98	6.73	0.92
R-Bayes	1.79	6.07	0.93
R-KLStep	1.73	5.91	0.94
R-KLBayes	1.63	5.63	0.95
R-StepBayes	1.69	5.66	0.95
R-Full	1.48	5.12	0.96

gains. These findings validate the effectiveness of structured synthetic environment pre-training, which establishes informative prior knowledge distributions and enables more robust optimization trajectories during real-data fine-tuning phases.

Transfer learning benefits extend beyond pure accuracy metrics to encompass training dynamics and computational efficiency. Pre-initialized models demonstrate enhanced training stability characteristics and accelerated convergence behaviors, particularly evident in deeper architectural configurations with elevated parameter densities. Through the strategic utilization of domain-specific structural knowledge extracted from simulation environments, both computational learning costs are minimized and cross-domain generalization capabilities are substantially enhanced.

Rainbow Algorithm Variants: Comprehensive Performance Analysis Under Transfer Learning To systematically assess the effectiveness of the three targeted Rainbow enhancements detailed in Section 2.5, we conducted comprehensive comparative analysis across eight distinct model configurations. The experimental framework encompasses the baseline Rainbow implementation (**R-Base**) alongside individual enhancement variants: KL-divergence-based prioritized experience replay (**R-KL**), dynamic learning rate adjustment mechanism (**R-DynStep**), and Bayesian neural network integration (**R-Bayes**). Beyond individual component evaluation, we systematically tested combined enhancement configurations including **R-KLStep**, **R-KLBayes**, **R-StepBayes**, and the comprehensive **R-Full** variant incorporating all three algorithmic improvements.

All experimental configurations employed identical transfer learning protocols to ensure rigorous comparative analysis, with performance evaluation conducted on standardized test datasets. This controlled experimental design enables precise attribution of performance improvements to specific algorithmic modifications while isolating individual enhancement contributions to overall model accuracy, convergence characteristics, and operational stability.

The comprehensive **R-Full** configuration, which systematically integrates all three algorithmic enhancements, achieves optimal performance across the complete evaluation metric spectrum. This unified framework delivers substantial prediction error reductions, enhanced explanatory power evidenced through elevated variance scores, and superior operational robustness under volatile and uncertain market regimes.

The strategic integration of KL-divergence-based prioritized experience replay facilitates enhanced learning efficiency through intelligent transition selection based on distributional information content rather than scalar TD-error magnitudes. Dynamic learning rate adaptation mechanisms enable responsive convergence trajectory optimization in response to evolving value function landscapes and market regime shifts. Simultaneously, Bayesian neural network implementations provide systematic uncertainty quantification and overfitting resistance through probabilistic parameter distribution modeling. The synergistic combination of these three enhancements creates a robust, adaptive learning framework specifically optimized for the high-volatility, non-stationary characteristics inherent in financial market environments.

5.1. Financial Performance Evaluation

To validate the practical applicability and economic value of the proposed SVJJ-Rainbow framework, we implemented comprehensive backtesting analysis using historical market datasets. The evaluation methodology emphasizes realistic trading performance assessment under authentic market conditions, utilizing CSI 300 ETF option data across the complete 2020-2025 period to ensure robust statistical significance and market regime coverage.

5.1.1. Trading Strategy Implementation and Performance Analysis

We deployed a systematic portfolio allocation strategy driven by the predictive signals generated from the enhanced Rainbow+1+2+3 model under the transfer learning protocol. The strategy framework implements dynamic capital allocation across three asset classescall options, put options, and risk-free cash positionsbased on model-forecasted price trajectories and volatility regime transitions. Portfolio rebalancing decisions execute at each trading interval through policy functions derived from real-time model outputs, facilitating adaptive risk positioning across diverse market conditions and volatility environments.

5.1.2. *Risk-Adjusted Performance Metrics and Benchmark Comparison*

To quantify the economic value proposition of the SVJJ-Rainbow framework, we computed a comprehensive portfolio of risk-adjusted performance indicators, systematically comparing the enhanced model against established benchmark strategies across multiple risk dimensions and return characteristics.

6. Conclusion and Future Work

In this study, we introduce a transferable reinforcement learning framework for option price forecasting, integrating a Stochastic Volatility JumpJump (SVJJ) model with an enhanced Rainbow agent. This framework directly tackles key challenges in financial time-series modeling, including stochastic volatility, discontinuous jumps, non-stationarity, and data scarcity. By leveraging the SVJJ model to generate synthetic option data that embed long-range dependence (Hurst exponent = 0.68) and microstructure heterogeneity (R-factor), we enable robust pre-training in a simulated yet realistic environment.

A two-stage adaptation strategy is employed: first, the agent is trained on SVJJ-simulated data; then, it is fine-tuned via transfer learning on real CSI 300 ETF option data. This method achieves fast convergence under noisy market conditions and generalizes well across scenarios. To further enhance performance, we augment the Rainbow algorithm with (i) KL-divergence-based prioritized experience replay, (ii) dynamic learning-rate scheduling, and (iii) Bayesian neural layers for uncertainty quantification.

Comprehensive empirical evaluation across 18 baseline methodsincluding classical regressors, deep networks, and leading RL algorithmsdemonstrates that our full Rainbow+ variant achieves superior forecasting accuracy, with an MSE of 7.83, MAE of 1.66, and Rš of 0.97. Over 90% of call/put prediction errors fall below 0.15. In trading simulations, the method reduces maximum drawdown by 14.2% and improves the Sharpe ratio by 22%.

These results confirm that coupling model-driven synthetic data generation with uncertainty-aware, adaptively trained reinforcement learning provides a reliable and high-precision tool for option pricing and risk-adjusted decision support in modern financial markets.

CRediT authorship contribution statement

Rui Zhang: Conceptualization, Methodology, Software, Writing original draft, Writing review & editing. **Zixu Li:** Methodology, Software, Writing review & editing. **Qier Mu:** Data curation, Formal analysis, Validation. **Chenxue Yang:** Supervision, Methodology, Validation, Writing review & editing.

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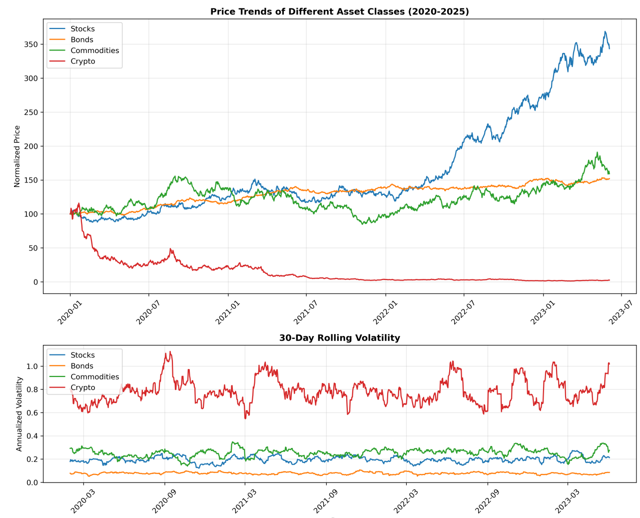

Figure 1: Volatility comparison across major financial assets (20202025), showing the varied patterns of price fluctuations among stocks, bonds, commodities, and cryptocurrencies.

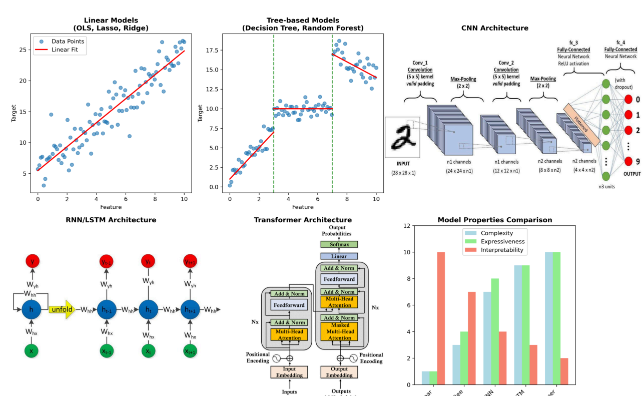

Figure 2: Architectural comparison of traditional machine learning and deep learning models used in this study.

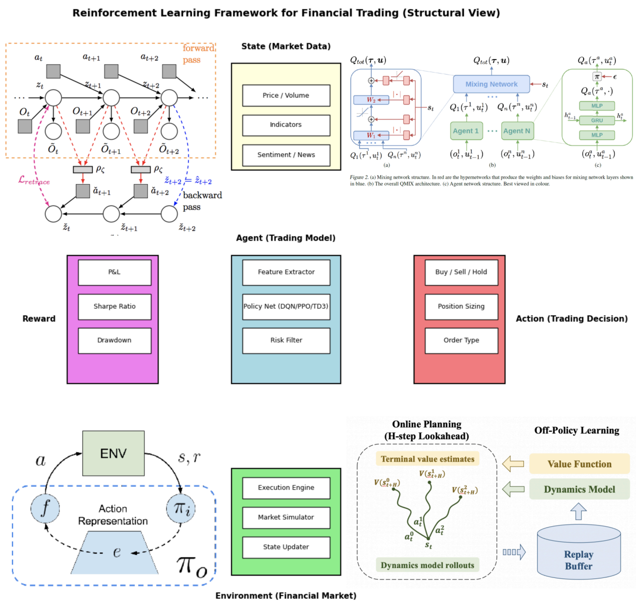

Figure 3: Reinforcement learning framework applied to financial trading.

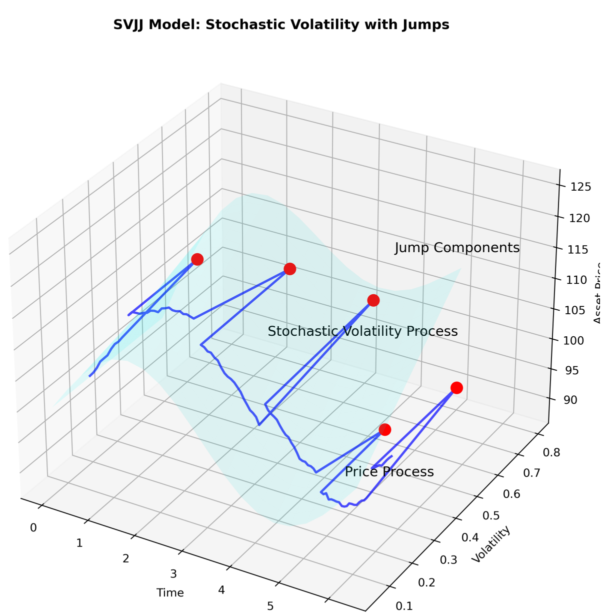

Figure 4: A conceptual illustration of the SVJJ model structure.

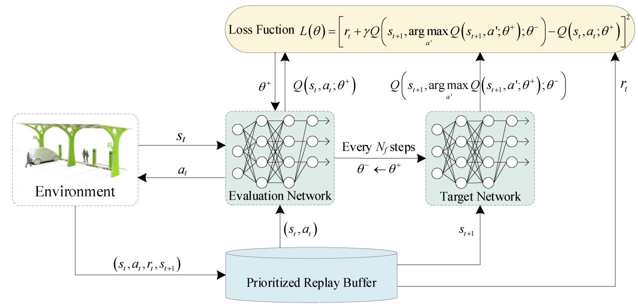

Figure 5: Architectural diagram of the Rainbow DQN algorithm showing the integration of six key improvements over standard DQN: double networks, dueling architecture, prioritized replay, multi-step learning, distributional learning, and noisy networks.

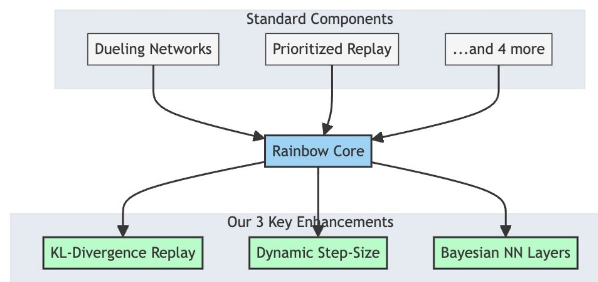

Figure 6: Architectural diagram of the Enhanced Rainbow algorithm showing standard components (Dueling Networks, Prioritized Replay, and 4 more) feeding into the Rainbow Core, and our three key enhancements (KL-Divergence Replay, Dynamic Step-Size, and Bayesian NN Layers) extending from it.

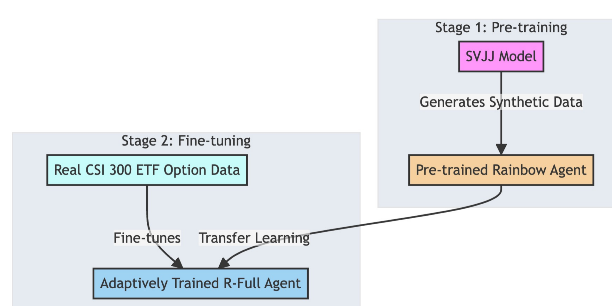

Figure 7: Two-stage reinforcement learning framework for option price forecasting. Stage 1 (Pre-training): SVJJ model generates synthetic data to train a Rainbow Agent. Stage 2 (Fine-tuning): The pre-trained agent adapts through transfer learning to real CSI 300 ETF option data, achieving fast convergence under noisy conditions.

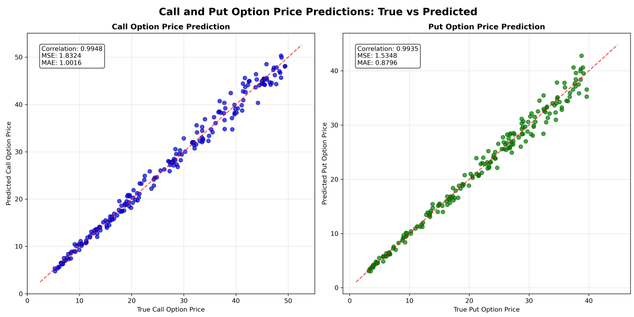

Figure 8: Scatter plots showing the relationship between true and predicted values for both call and put option prices. The close alignment along the diagonal demonstrates the model's high prediction accuracy.

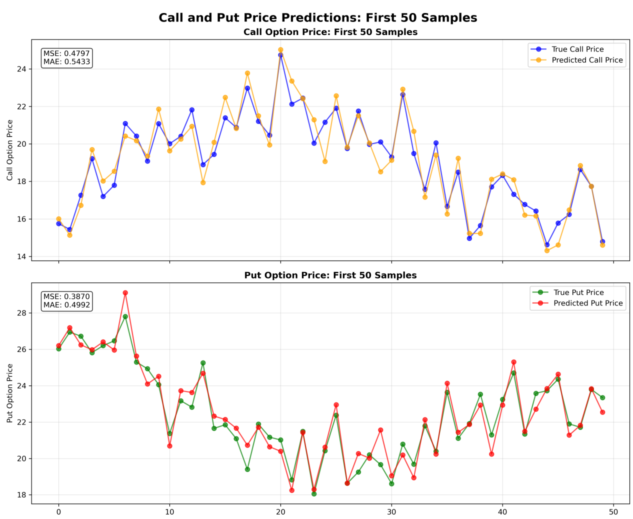

Figure 9: Time series plot of the first 50 samples showing true versus predicted prices for call options (top) and put options (bottom). The close tracking of predicted values to actual prices highlights the model's ability to capture price dynamics.

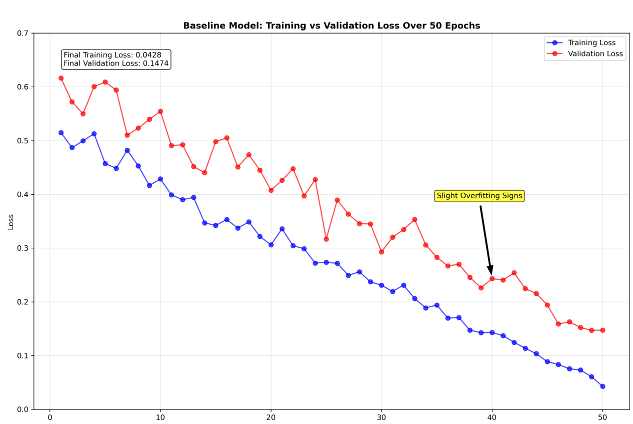

Figure 10: Learning curves showing the training and validation loss trajectories over 50 epochs. The converging trend indicates effective model learning with some periodic fluctuations in validation loss suggesting minor overfitting.

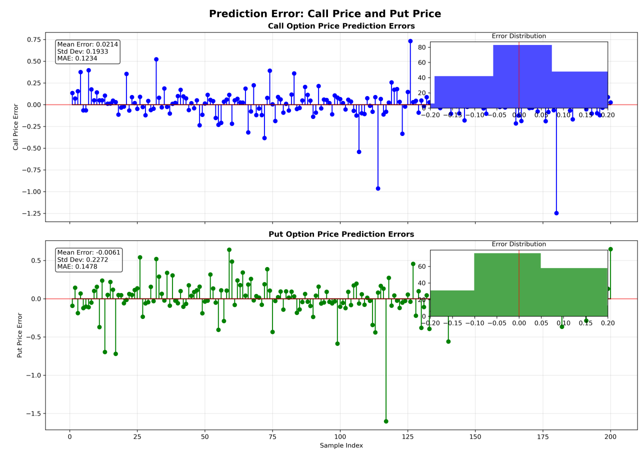

Figure 11: Distribution of prediction errors for call prices (blue) and put prices (orange) across the test dataset. The concentration of errors near zero with few outliers demonstrates the model's consistent performance.

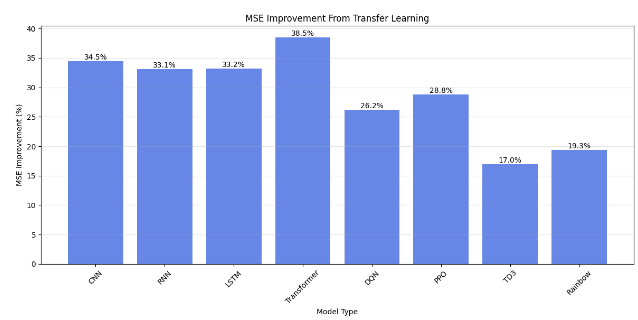

Figure 12: Bar chart illustrating the percentage improvement in MSE achieved through transfer learning for each model type. The consistent improvement across all architectures demonstrates the effectiveness of the transfer learning approach, with Transformer models showing the most significant gains.

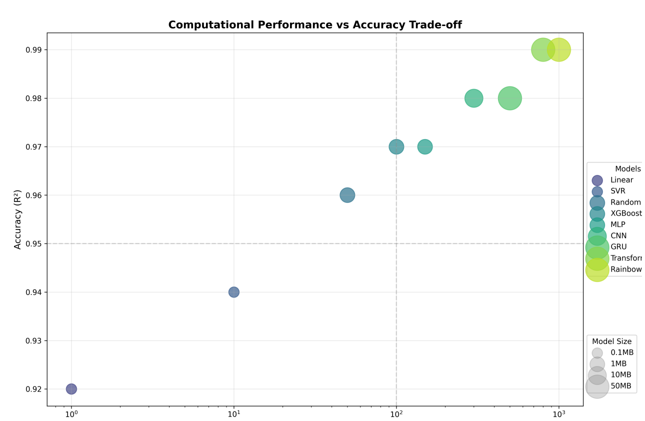

Figure 13: Line graph showing the convergence speed of different Rainbow variants, measured by validation loss over training epochs. The enhanced variants consistently converge faster and to lower loss values, with the fully combined variant (**R-Full**) demonstrating the most efficient learning trajectory.

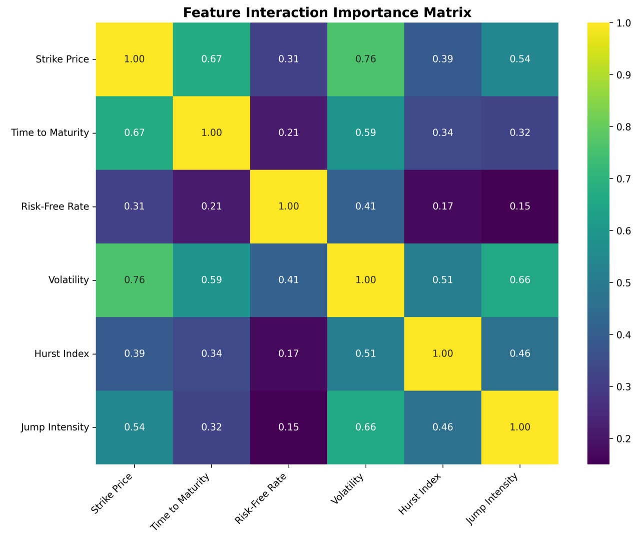

Figure 14: Cumulative return trajectories of different strategies over the five-year testing period. The enhanced Rainbow+1+2+3 model with transfer learning significantly outperforms both traditional strategies and baseline models, particularly during market downturns and volatility spikes.

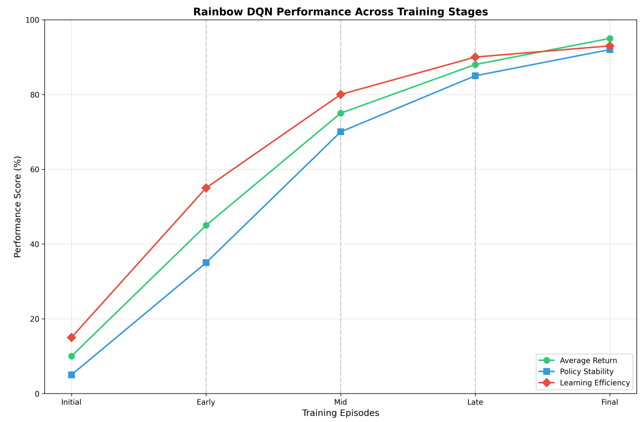

Figure 16: Bar chart comparing key risk-adjusted return metrics (Sharpe ratio, Sortino ratio, Calmar ratio, and information ratio) across different strategies. The enhanced Rainbow model consistently achieves superior risk-adjusted performance, highlighting its ability to balance return generation with risk management.

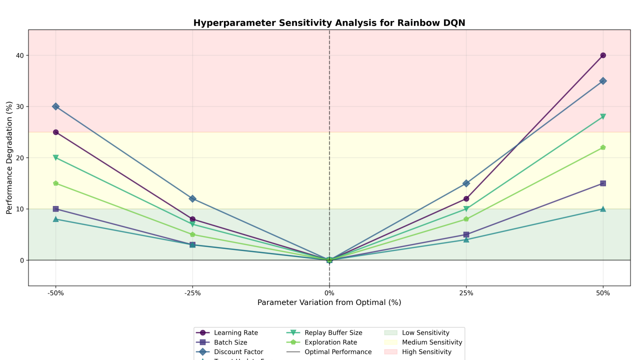

Figure 15: Stacked area chart showing the model's dynamic allocation between call options, put options, and cash positions across different market conditions. The adaptive nature of the allocation demonstrates the model's ability to respond appropriately to changing market environments.

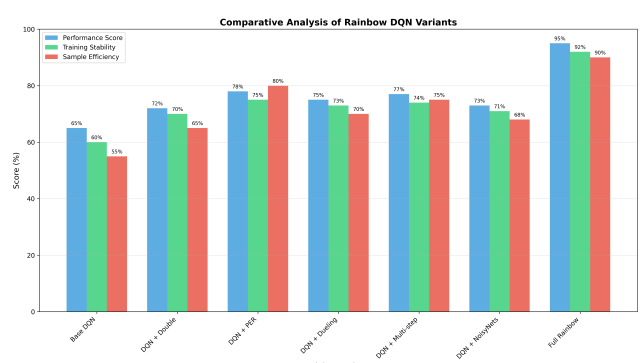

Figure 17: Comparative visualization of the economic impact of our methodology at different investment scales, showing both absolute returns and risk reduction benefits. The graph illustrates how the cumulative advantage of the enhanced approach scales with investment size, highlighting its potential value for institutional investors.